



BUC
BADR UNIVERSITY IN CAIRO
جامعة بدر بالقاهرة

Academic year: 2024/2025
Semester: 2nd semester
Exam: Final Term Exam
Code: BAS 031
Date: 28/5/2025

Faculty of Engineering and
Technology
Basic Science Department
Course level: 0
Allowed Time: 3 Hour

Course Title: Mechanics (1)-Statics

Number of Exam Questions: 4

Max Exam Grade: 50 marks

Course Offered By

☒ FRE ☐ STR ☐ ART ☐ ENR ☐ MET ☐ CMP ☐ COM ☐ ENG ☐ HUM

Exam Instructions (plagiarized answers will receive zero marks)

1. Type of Exam: ☒ Closed-Book ☐ Open-Book ☐ Charts and Formula Sheets Only
2. Write your answers in: ☒ Same Booklet ☐ Separate Booklet ☐ Drawing Sheet
3. Attempt ALL questions

Question-1 (13 marks)

A transmission tower is held by three guy wires attached to a pin at A and anchored by bolts at B, C, and D as shown in Figure 1. If the tension in wire AB is 630 lb, determine the vertical force P exerted by the tower on wires at A.

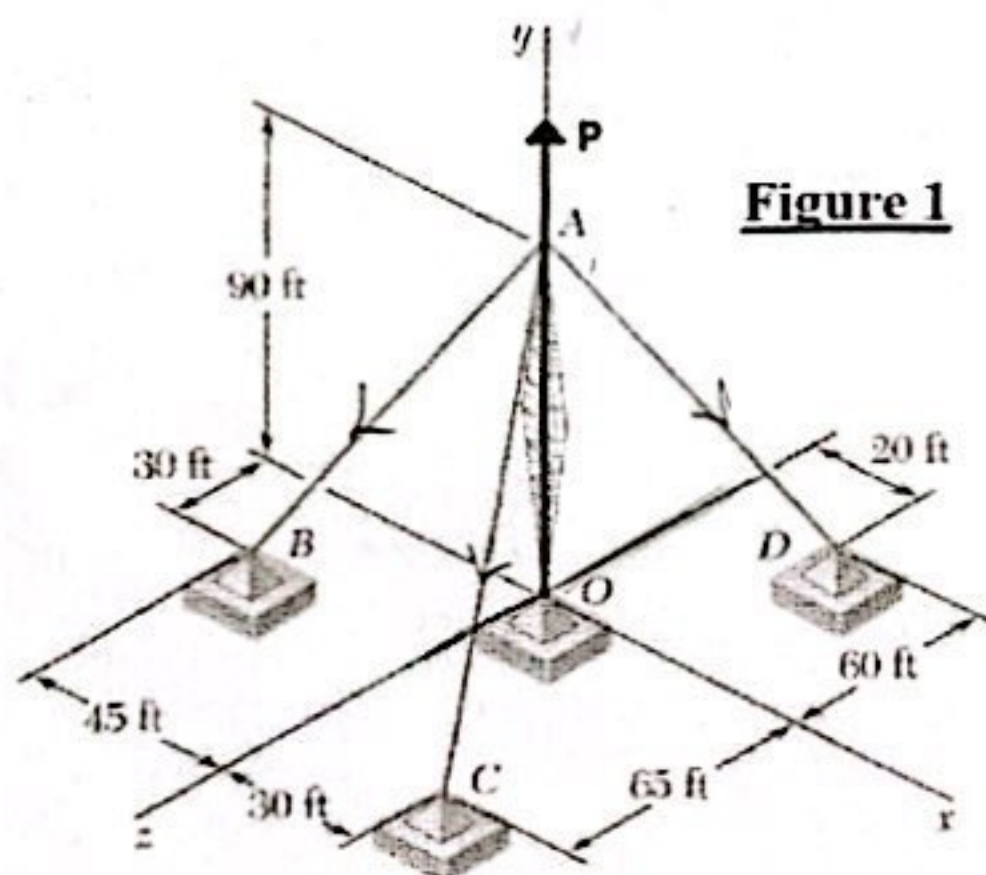


Figure 1

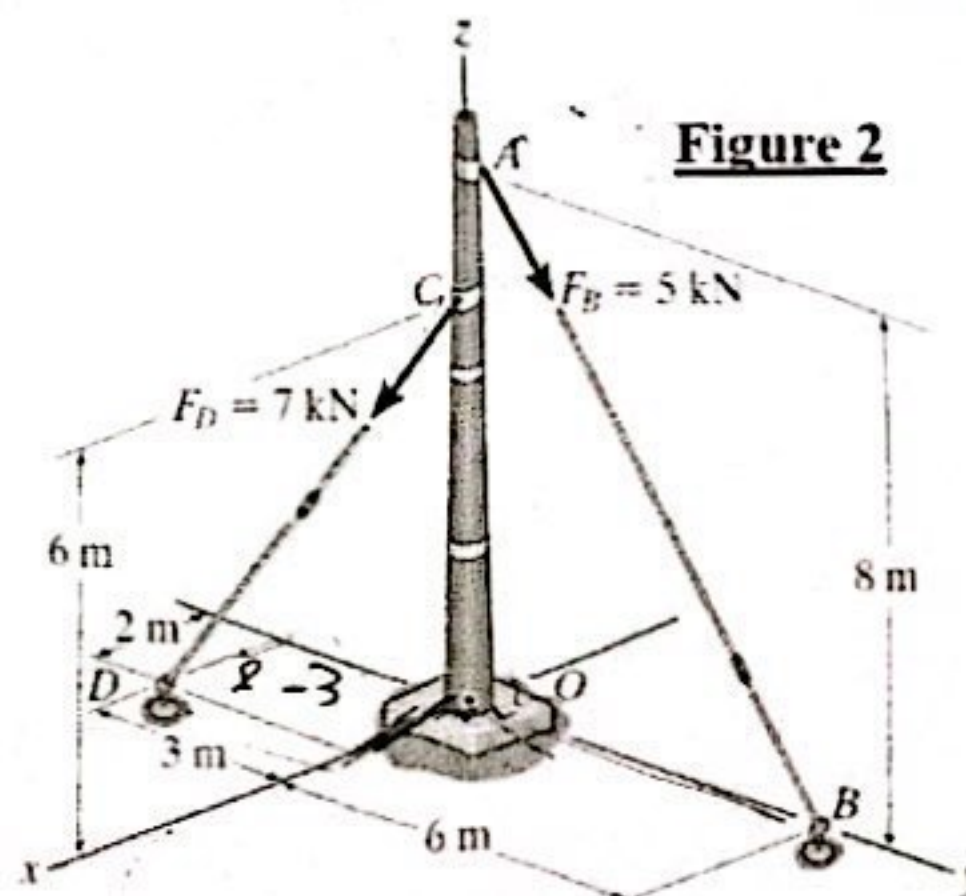


Figure 2

Question-2 (12 marks)

Determine the moment about point O of the two forces acting on the post shown in Figure 2.



Question-3 (13 marks)

Determine the force components acting on the ball-and-socket at D, the reaction at the roller C and the tension on the cord AB needed for equilibrium of the quarter circular plate shown in Figure 3. Neglect the mass of the plate.

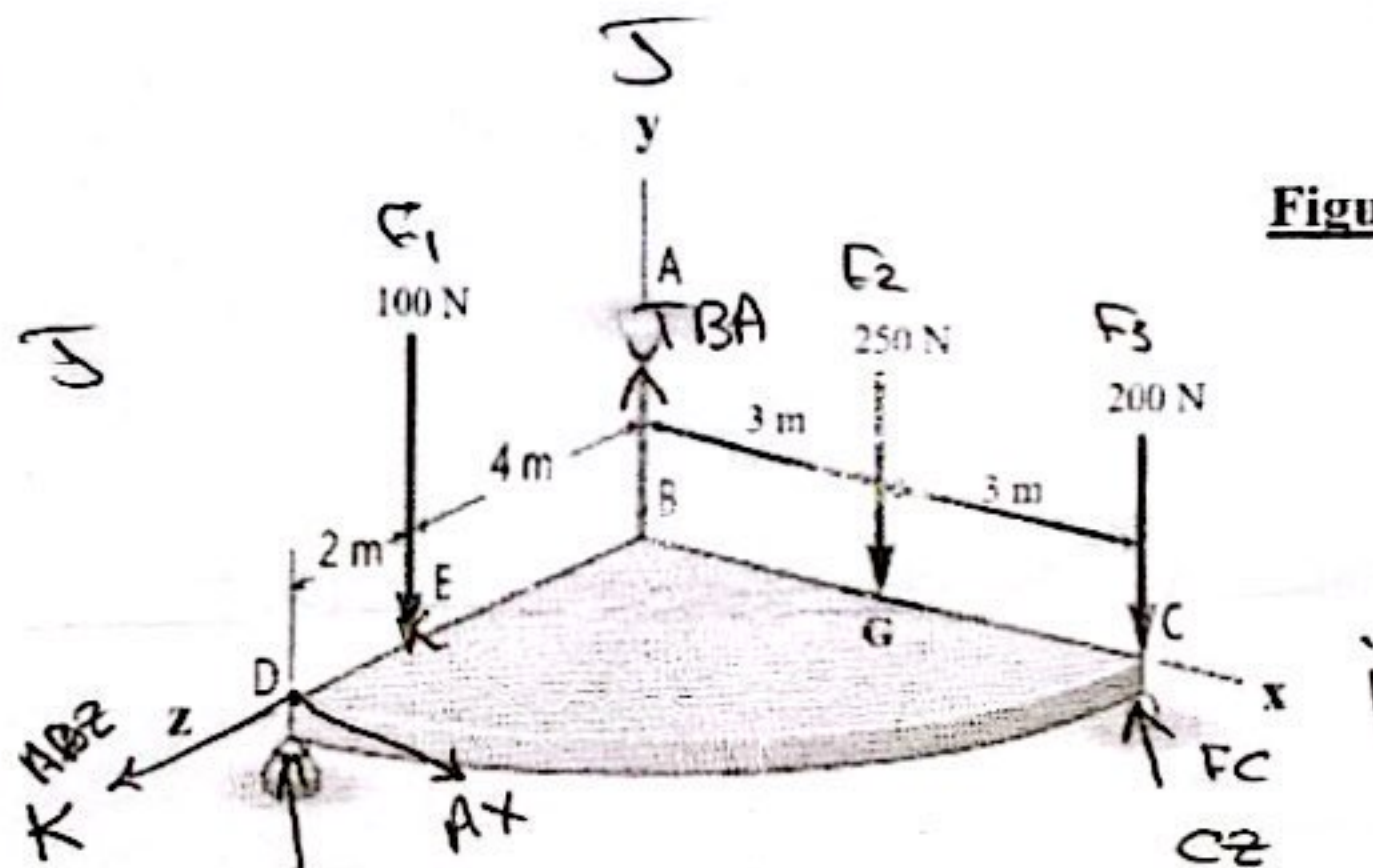


Figure 3

Question-4 (12 marks)

For the plane area shown in Figure (4), determine the location of the centroid.

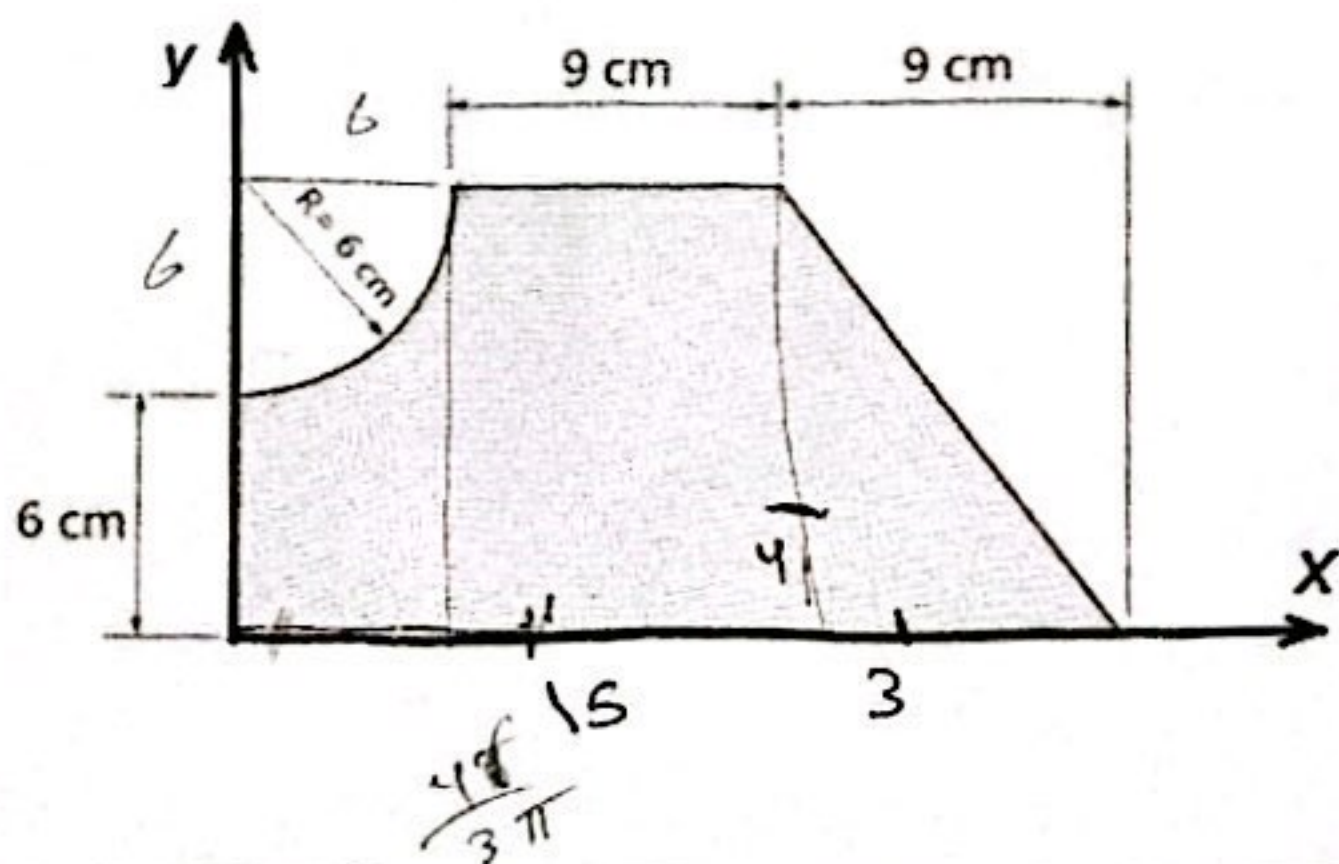


Figure 4

End the questions

With best wishes

Dr. Ahmed El-Desouky



Faculty of Engineering and Technology
Basic Science Department
Course level: 0
Allowed Time: 1 Hour

☒ *FRE* ☐ *STR* ☐ *ART* ☐ *ENR* ☐ *MET* ☐ *CMP* ☐ *COM* ☐ *ENG* ☐ *HUM*



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Academic year: 2024/2025
Semester: 1st semester
Exam: Final Term Exam
Code: BAS 031
Date: 11/1/2025

Faculty of Engineering and
Technology
Basic Science Department
Course level: 0
Allowed Time: 3 Hour

Course Title: Mechanics (1) Statics

Number of Exam Questions: 4

Max Exam Grade: 50 marks

Course Offered By

☒ FRE ☐ STR ☐ ART ☐ ENR ☐ MET ☐ CMP ☐ COM ☐ ENG ☐ HUM

Exam Instructions (plagiarized answers will receive zero marks)

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3. Attempt ALL questions

Question-1 (10 marks)

A 2.6 lb force is applied at Point D of the bent post shown in Figure 1. Determine the moment of that force about the center A of the base section.

Figure 1

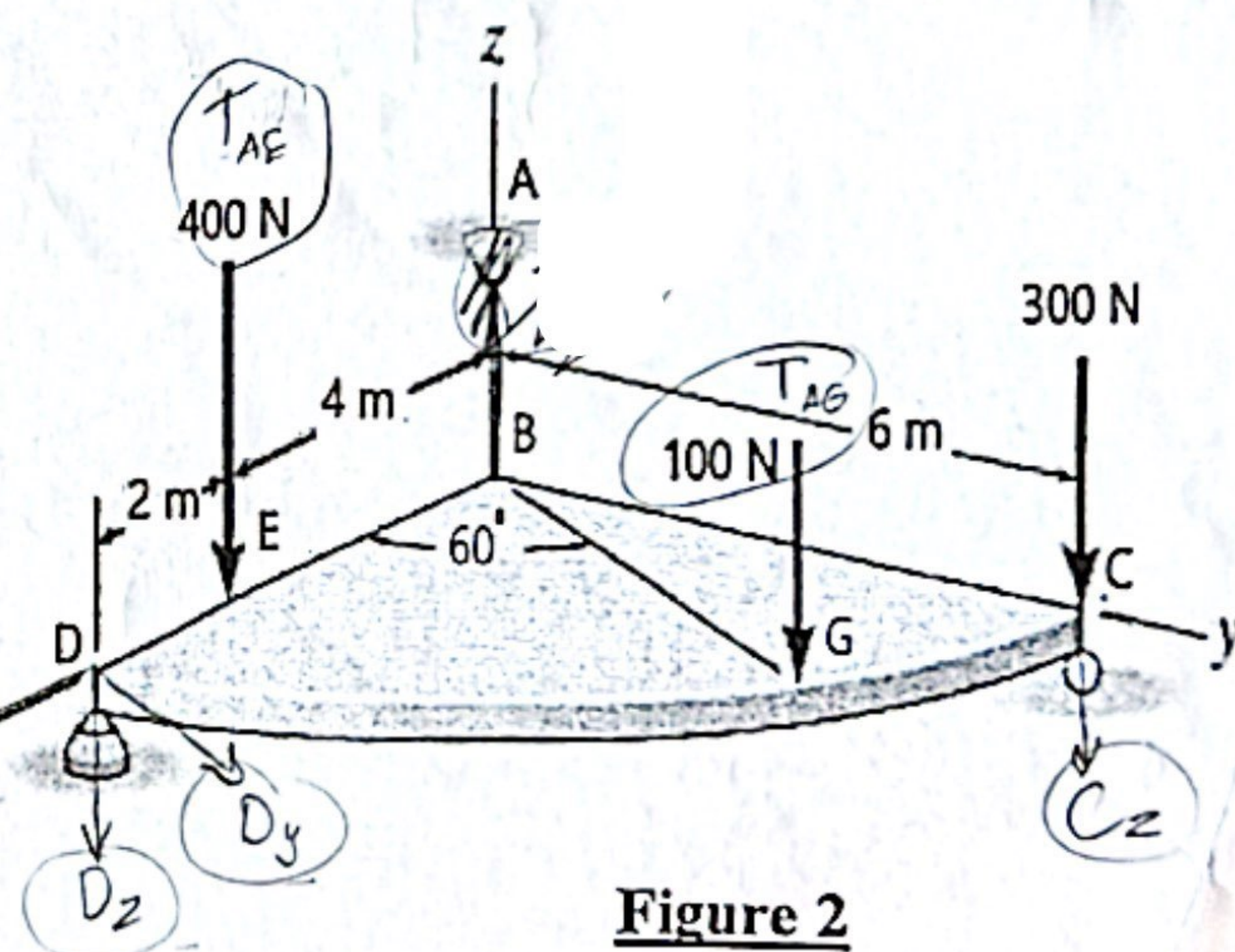
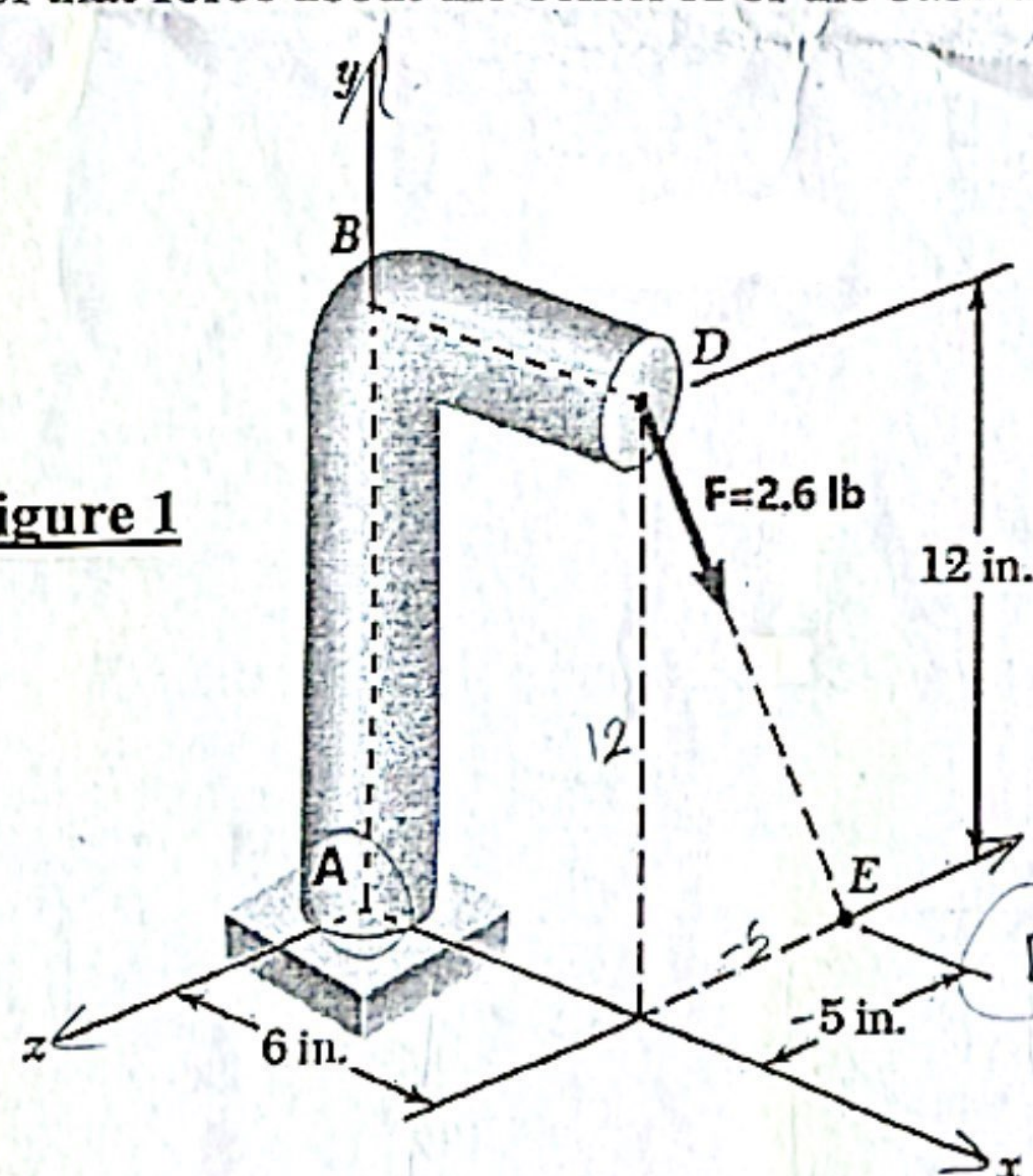


Figure 2

Question-2 (15 marks)

Determine the force components acting on the ball-and-socket at D, the reaction at the roller C and the tension on the cord AB needed for equilibrium of the quarter circular plate shown in Figure (2). Neglect the mass of the plate.



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Basic Science Department
Course level: 0
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Question-3 (12 marks)

For the truss and load shown in Figure (3), Determine;

- 1) the reactions at rocker support E and hinge support A,
- 2) the forces in members BC, CE, and ED, and indicate whether the members are in tension or compression using the method of sections.

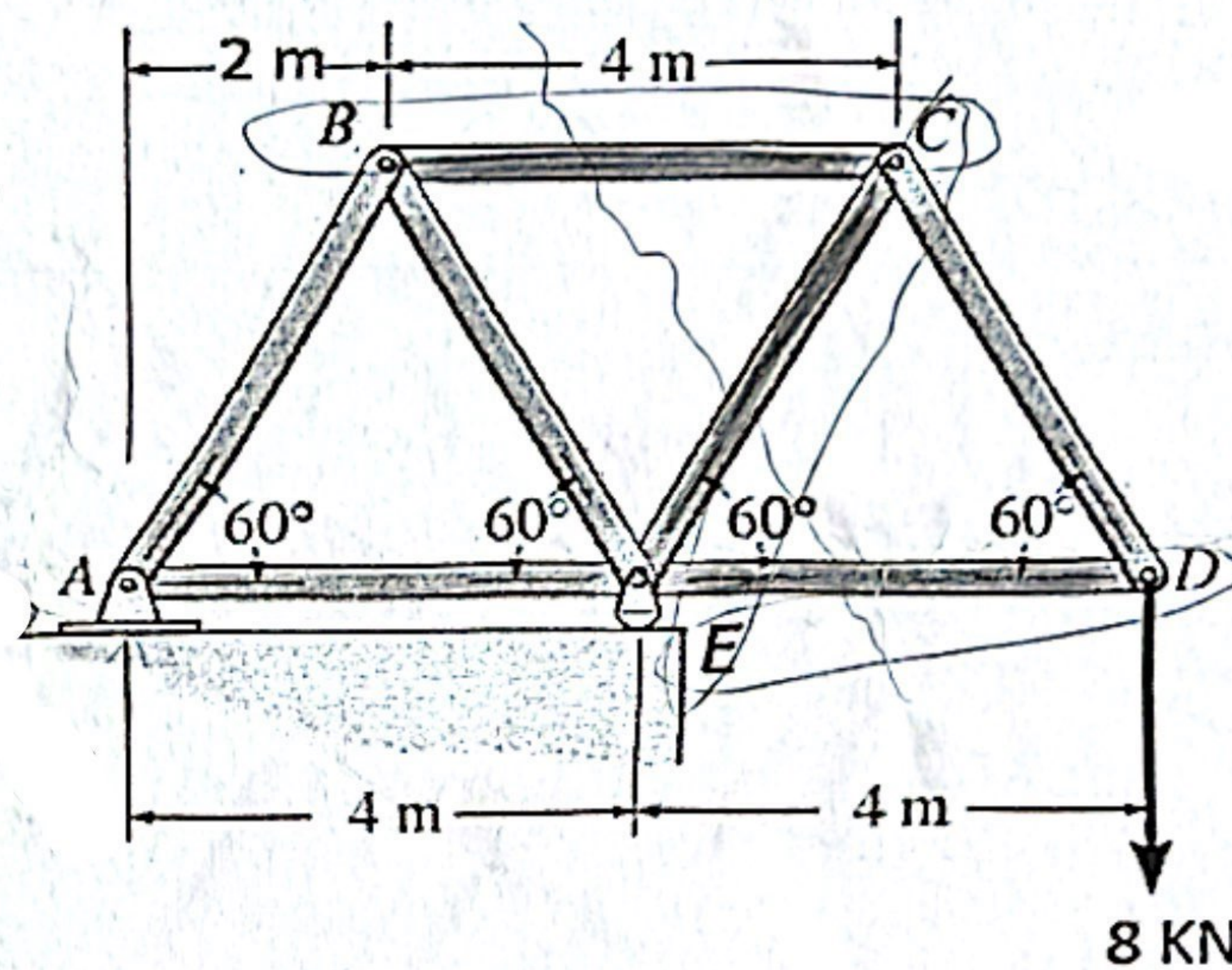
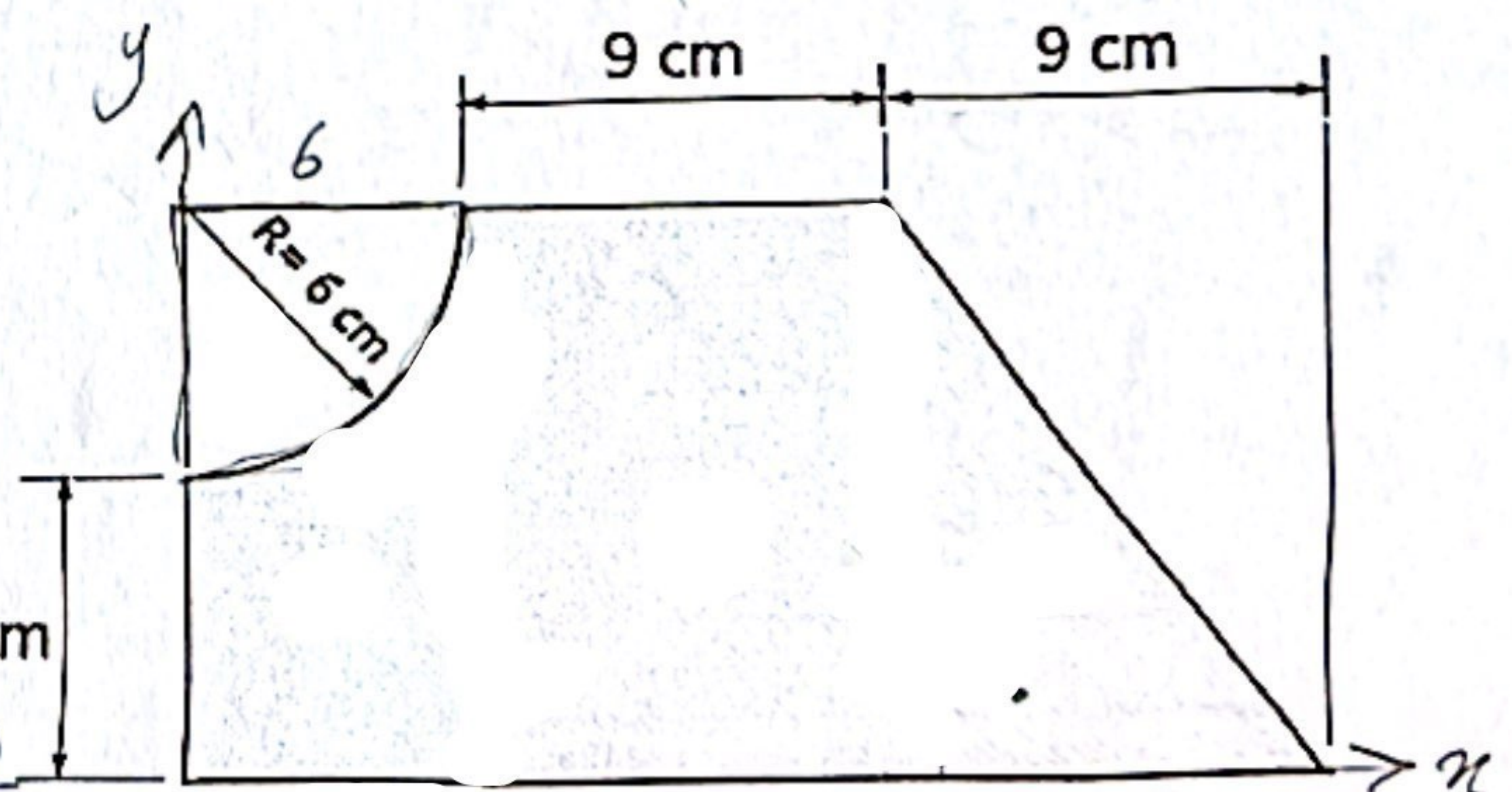


Figure 3

Question-4 (13 marks)

For the plane area shown in Figure (4), determine the location of the centroid.

Figure 4



$$\bar{x} = \frac{\sum x \cdot \text{Area}}{\sum \text{Area}}$$

End the questions

$$\bar{y} = \frac{\sum y \cdot \text{Area}}{\sum \text{Area}}$$

With best wishes

Dr. Ahmed El-Desouky



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جامعة بدر بالقاهرة

Academic year: 2023/20234
Semester: 1st semester
Exam: Midterm Exam
Code: BAS 031
Date: 18/11/2024

Faculty of Engineering and
Technology
Basic Science Department
Course level: 0
Allowed Time: 1 Hour

Course Title: Mechanics I

Number of Exam Questions: 2

Max Exam Grade: 20 marks

Course Offered By

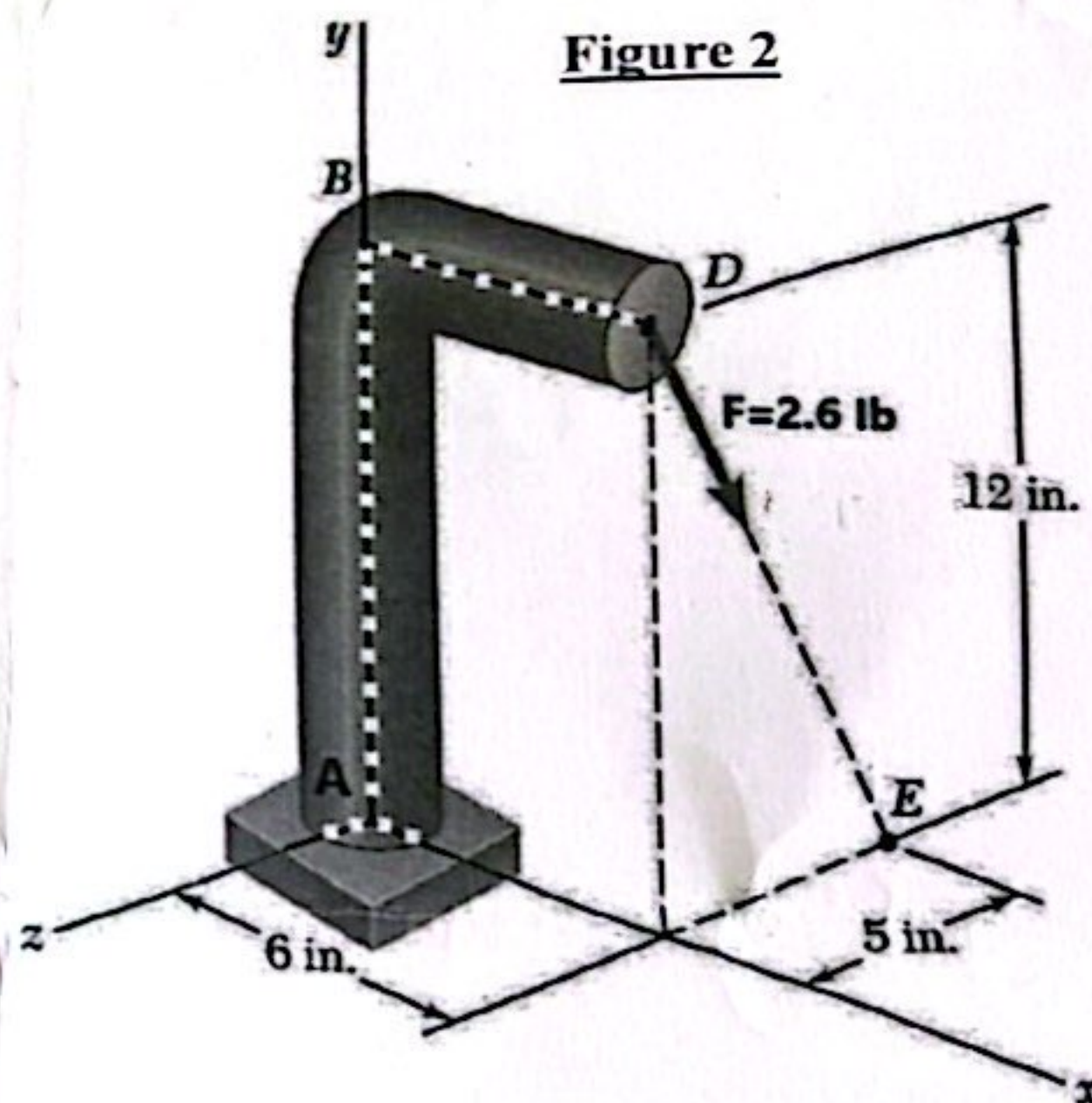
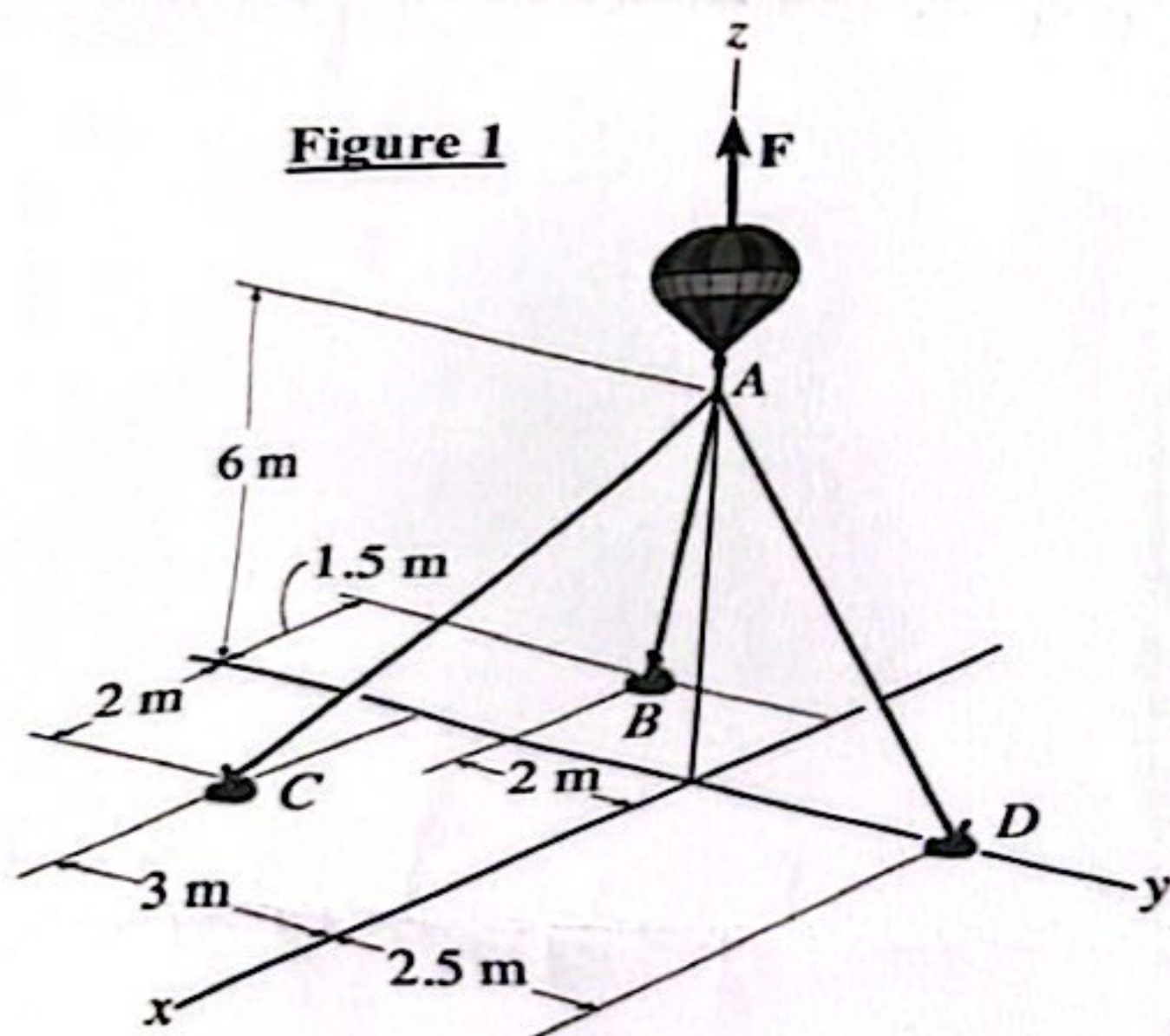
☒ FRE ☐ STR ☐ ART ☐ ENR ☐ MET ☐ CMP ☐ COM ☐ ENG ☐ HUM

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Question-1 (10 marks)

If the balloon is subjected to a net uplift force of $F = 800$ N as shown in Figure 1, determine the tension developed in each rope.



Question-2 (10 marks)

A 2.6 lb force is applied at Point D of the cast iron post shown in Figure 2. Determine the moment of that force about the center A of the base section.

With best wishes

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Academic year: 2023/20234
Semester: 2nd semester
Exam: Final Term Exam
Code: BAS 031
Date: 4/6/2024

Faculty of Engineering and
Technology
Basic Science Department
Course level: 0
Allowed Time: 3 Hour

Course Title: Mechanics (I) Statics

Number of Exam Questions: 5

Max Exam Grade: 50 marks

Course Offered By

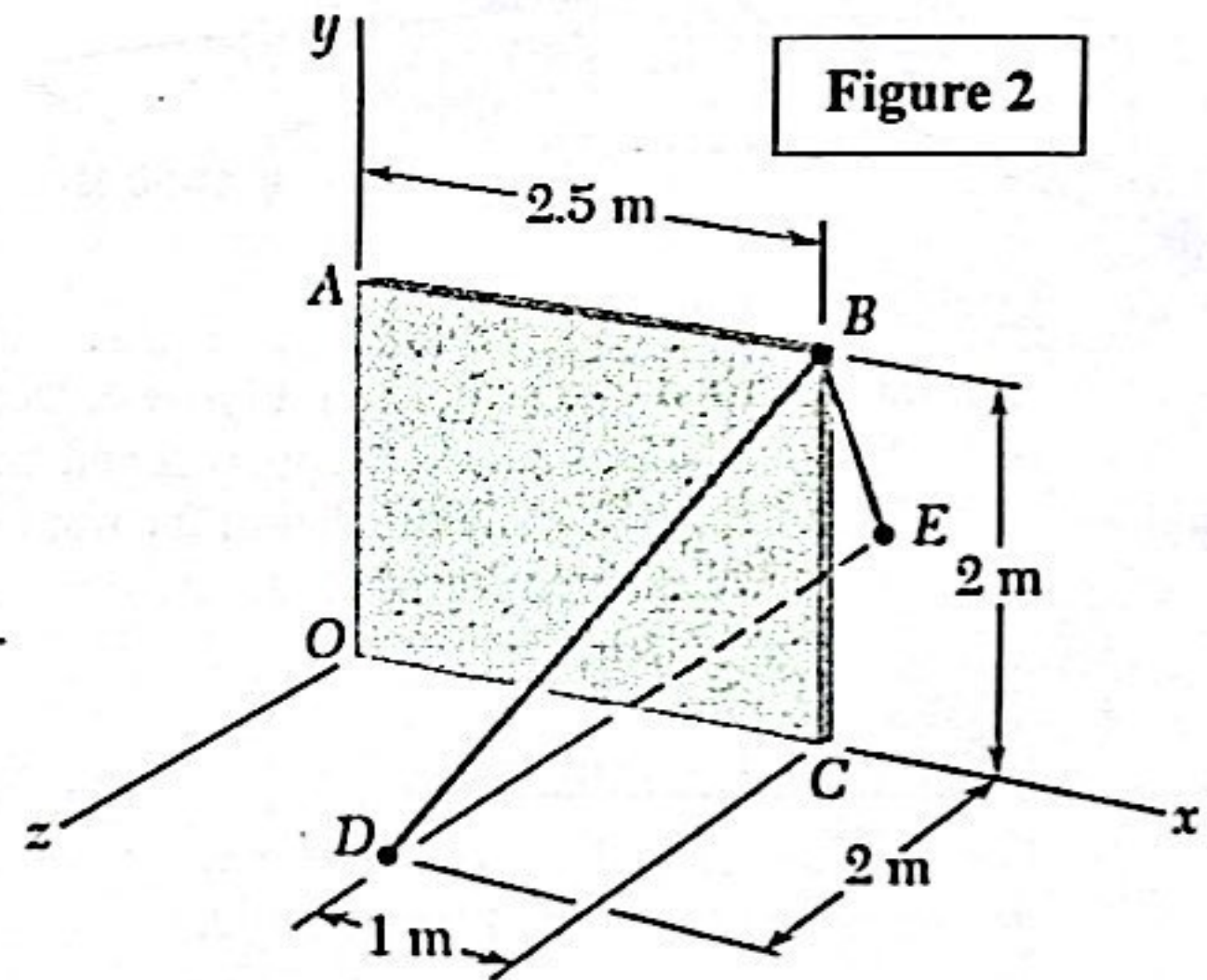
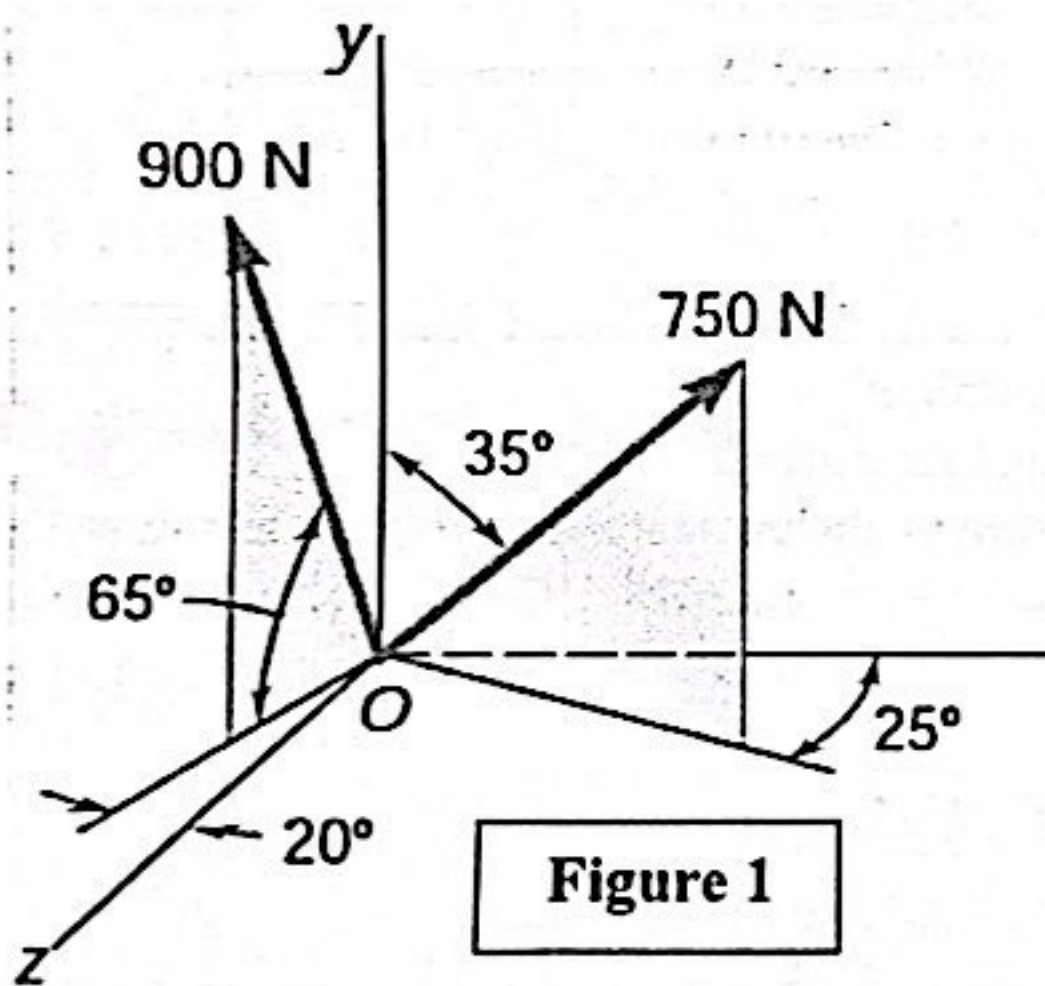
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Exam Instructions (plagiarized answers will receive zero marks)

1. Type of Exam: ☒ Closed-Book ☐ Open-Book ☐ Charts and Formula Sheets Only
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Question 1 (5 marks)

Determine the resultant of the two forces shown in Figure 1.



Question 2 (10 marks)

The concrete wall section is temporarily held by two cables as shown in Figure 2. Knowing that the tension in cable BD is 6000 N, determine the moment about point A of the force exerted by the cable BD at point B.



Question 3 (15 marks)

A 10-m boom is acted upon by the 1000-N force shown in Figure 3. Determine the tension in each cable and the reaction at the ball-and-socket joint at A.

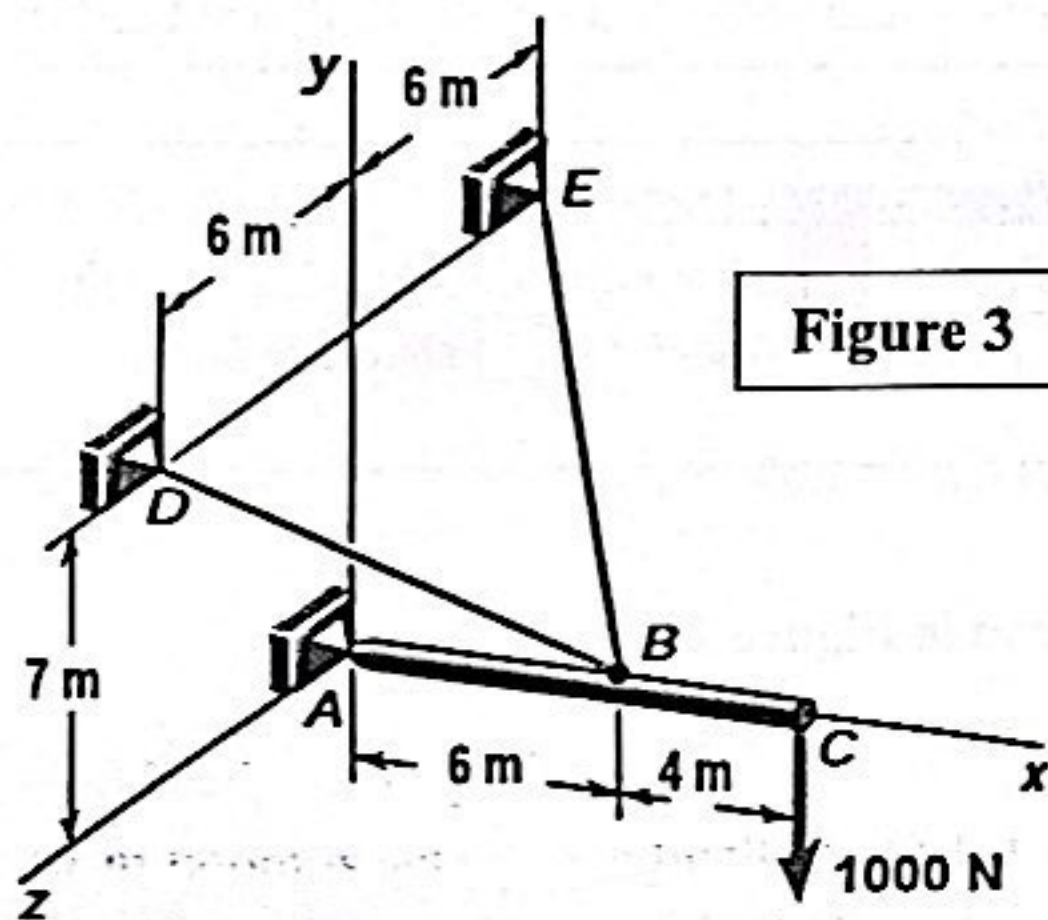


Figure 3

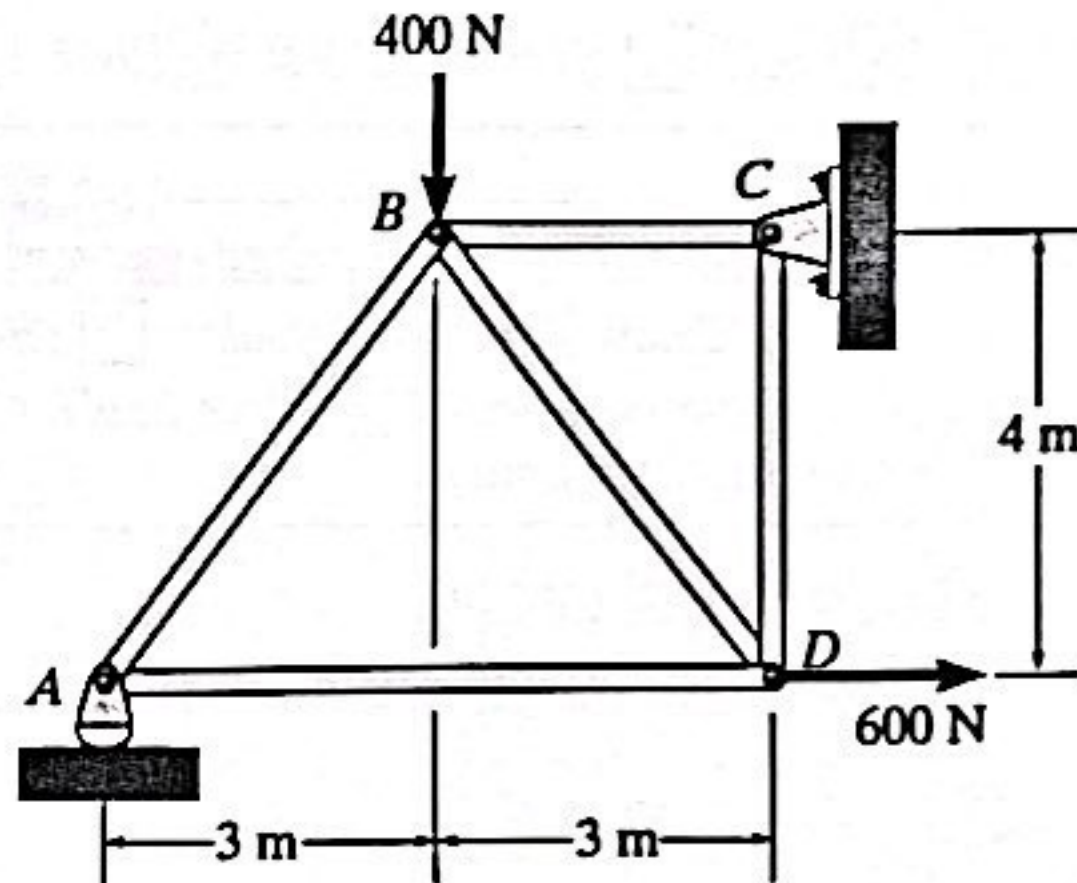


Figure 4

Question 4 (10 marks)

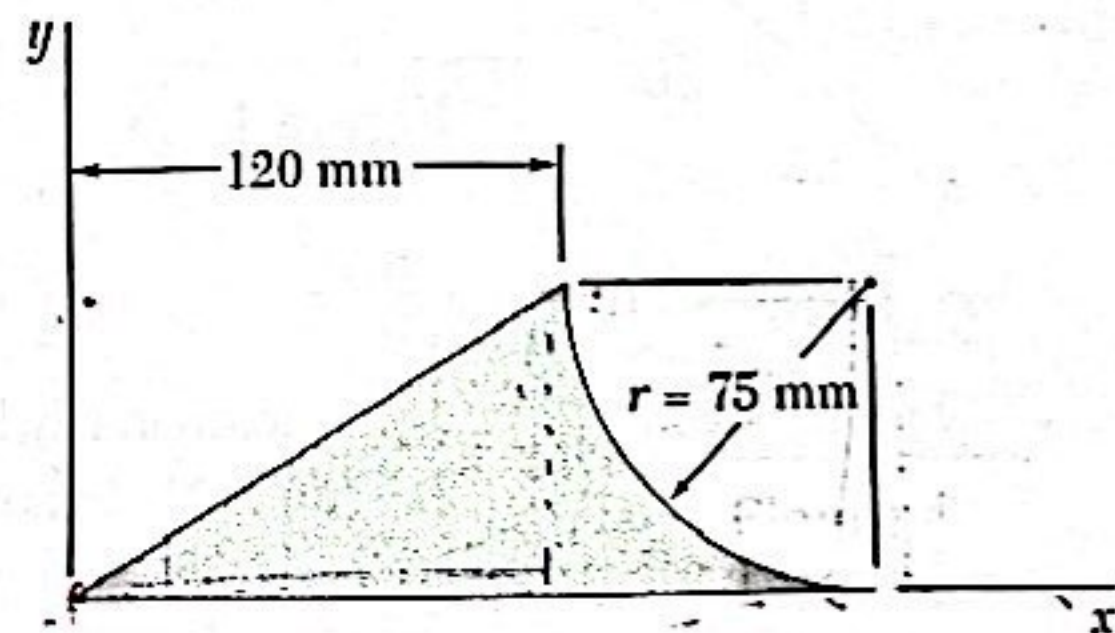
For the truss and loads shown in Figure 4. Determine:

- the reactions at roller support A and hinge support C,
- the force in each member of the truss shown and indicate whether the members are in tension or compression.

Question 4 (10 marks)

For the plane area shown in Figure 5, determine the location of the centroid.

Figure 5



End the questions

With best wishes

Dr. Ahmed El-Desouky



Course Title: Mechanics (1)

Number of Exam Questions: 4

Max Exam Grade: 50 marks

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Question-1 (10 marks)

A 1000 N cylinder is hung by means of two cables AB and AC, which are attached to the top of a vertical wall as shown in figure (1). A horizontal force P perpendicular to the wall holds the cylinder in the position shown. Determine the magnitude of the force P and the tension in each cable.

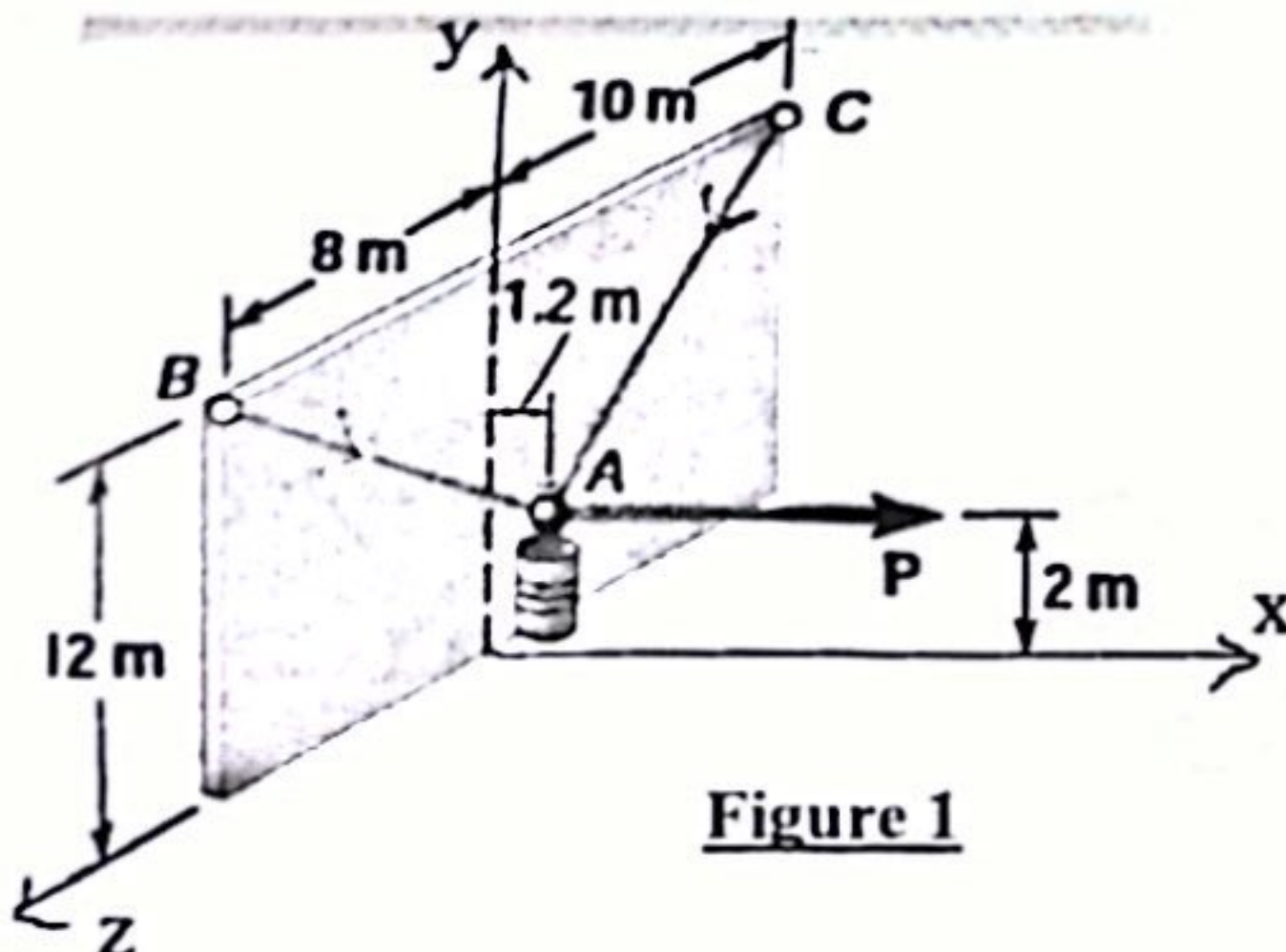


Figure 1

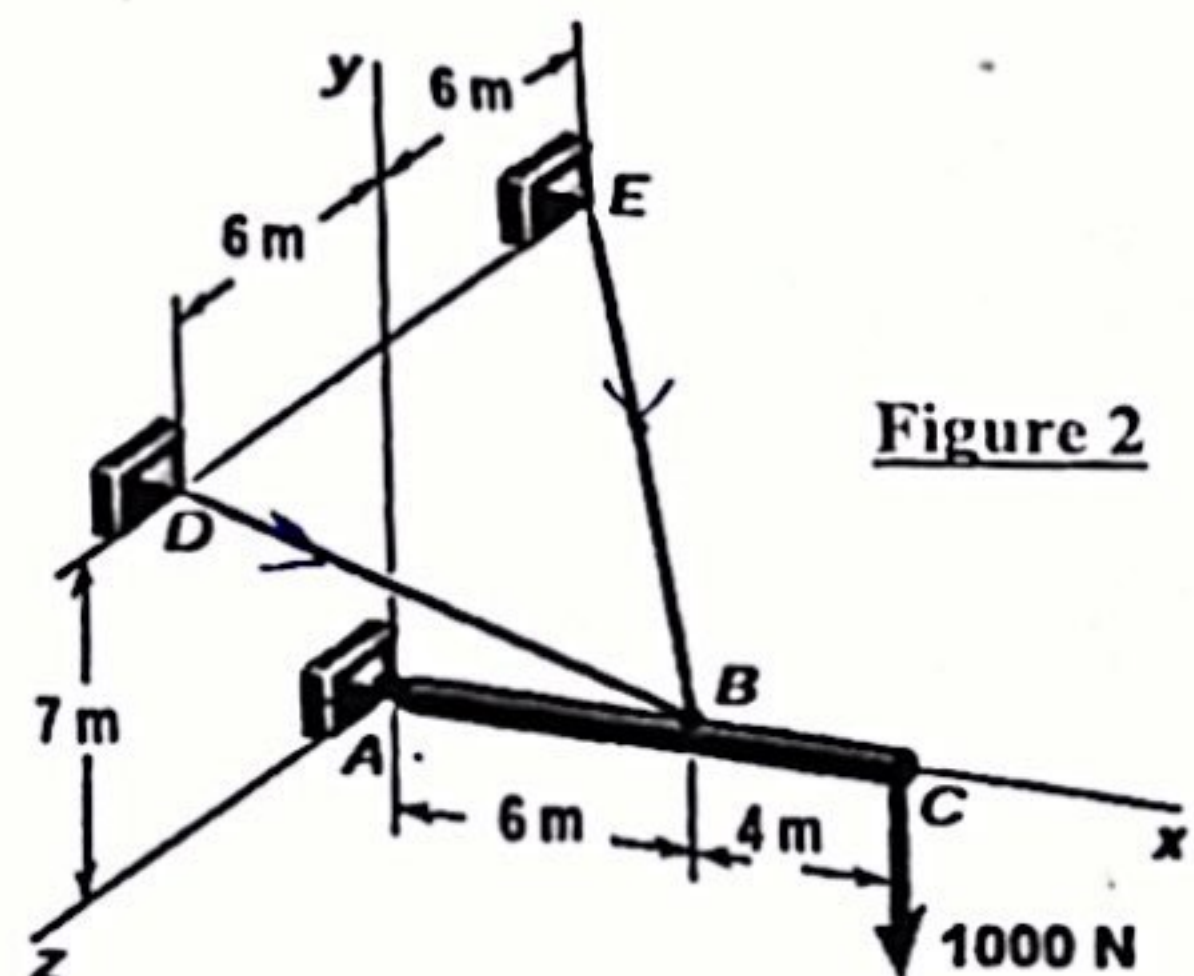


Figure 2

Question-2 (15 marks)

A 10-m boom is acted upon by the 1000-N force shown in figure (2). Determine the tension in each cable and the reaction at the ball-and-socket joint at A.



Question-3 (15 marks)

a) (5 marks): Determine the zero-force members in truss and loads shown in figure (3).

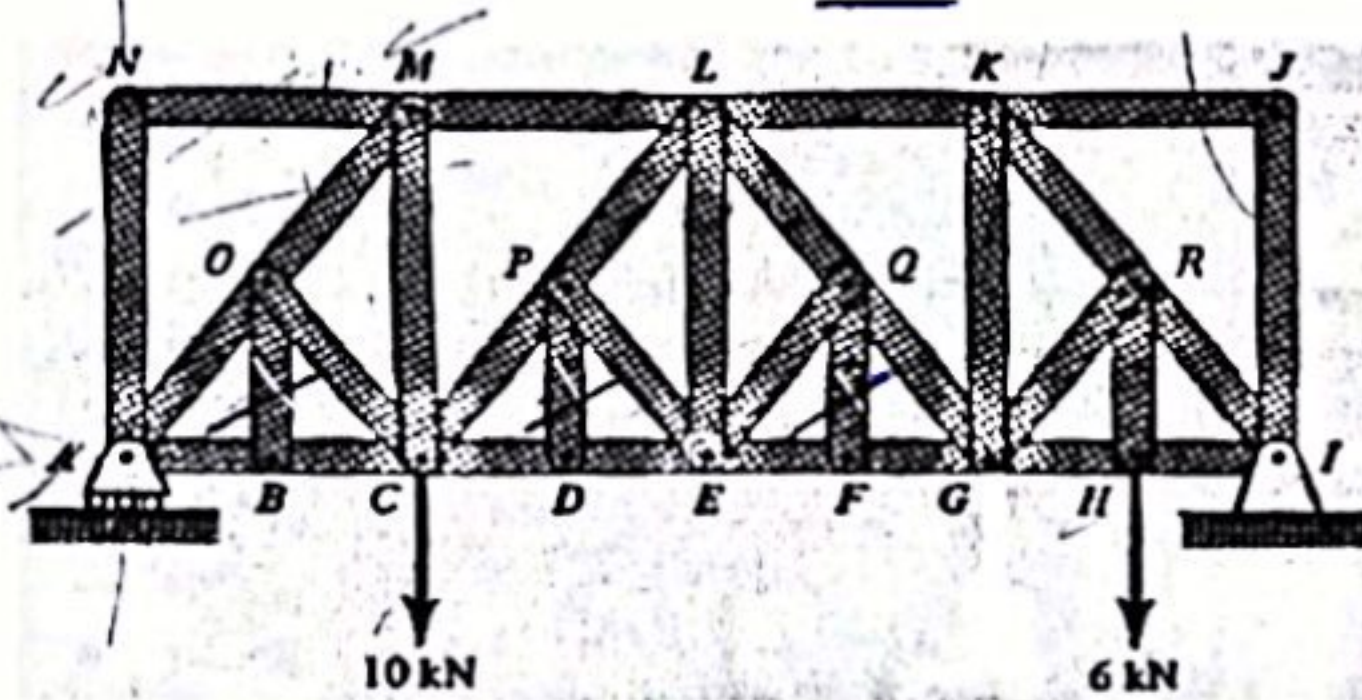


Figure 3

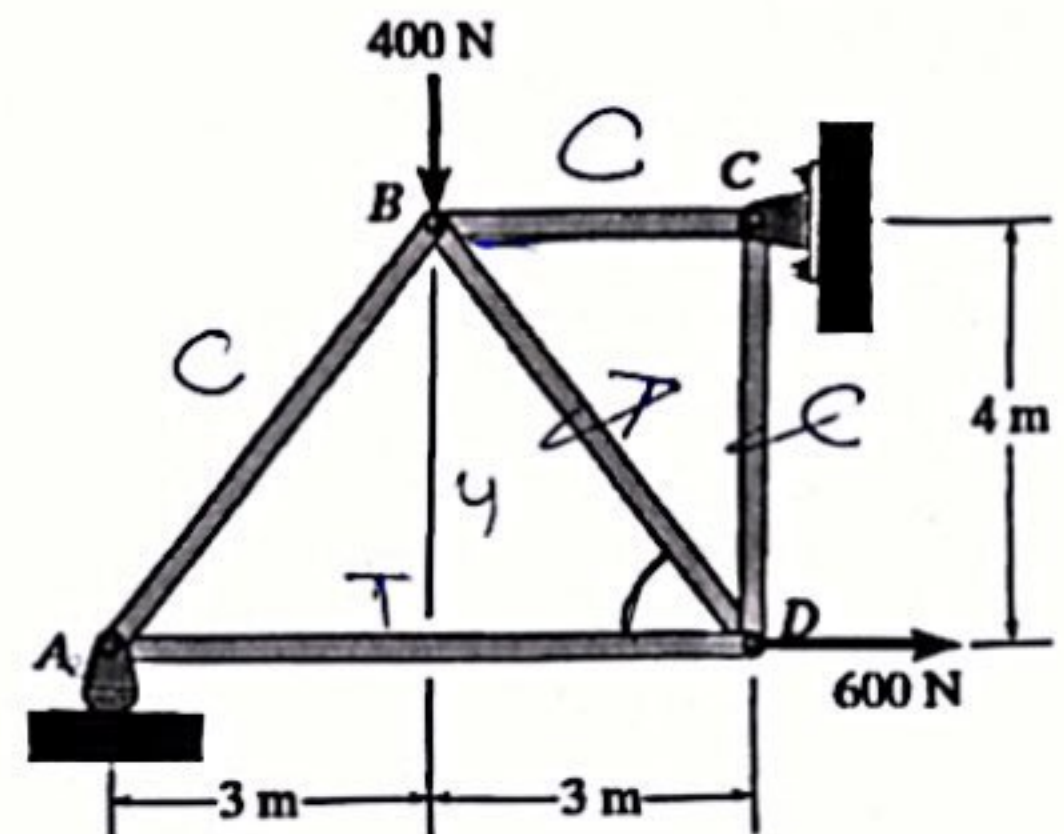


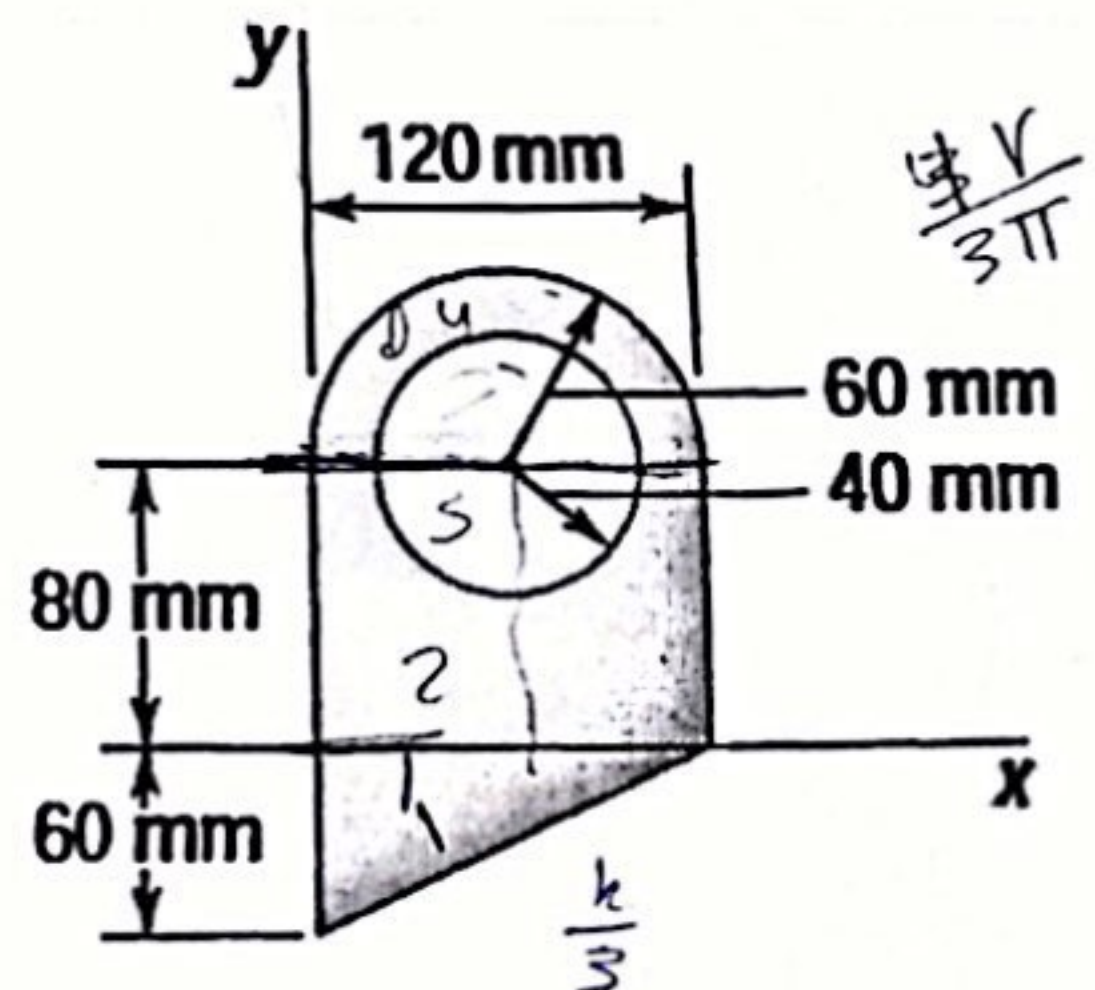
Figure 4

- a) (10 marks): For the truss and loads shown in figure (4). Determine,
- 1) the reactions at roller support A and hinge support C,
 - 2) the force in each member of the truss shown and indicate whether the members are in tension or compression.

Question-4 (10 marks)

For the plane area shown in figure (5), determine the location of the centroid.

Figure 5



End the questions

With best wishes

Dr. Ahmed El-Desouky